

```
In [1]: # import the packages
# read the data
# divide cat and num columns

import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns

visa_df=pd.read_csv(r'C:\Users\omkar\OneDrive\Documents\Gen_AI\Data_files\Visada
```

```
In [3]: cat=visa_df.select_dtypes(include='object').columns
num=visa_df.select_dtypes(exclude='object').columns
```

```
In [5]: num
```

```
Out[5]: Index(['no_of_employees', 'yr_of_estab', 'prevailing_wage'], dtype='object')
```

```
In [7]: wage_data=visa_df['prevailing_wage'] # selection of one column
```

```
In [ ]: - count

- min

- 25p

- mean

- median

- 50p

- 75p

- max

- std
```

```
In [13]: count=len(wage_data)

Min=wage_data.min()
Min=np.min(wage_data)

P_25= np.percentile(wage_data,25) # 1 to 100
P_25=np.quantile(wage_data,0.25) # 0 to 1

Mean=wage_data.mean()
Mean= np.mean(wage_data)

Median=wage_data.median()
Median= np.median(wage_data)

P_50= np.percentile(wage_data,50) # 1 to 100
P_50=np.quantile(wage_data,0.5) # 0 to 1

P_75= np.percentile(wage_data,75) # 1 to 100
P_75=np.quantile(wage_data,0.75) # 0 to 1
```

```

Max=wage_data.max()
Max=np.max(wage_data)

SD=wage_data.std()
SD=np.std(wage_data)

```

```

In [21]: l=[count,Min,P_25,Mean,Median,P_50,P_75,Max,SD]
idx=['Count','Min','25P','Mean','Median','50P','75P','Max','STD']
pd.DataFrame(l,
             columns=['prevailing_wage'],
             index=idx)

```

Out[21]:

prevailing_wage	
Count	25480.000000
Min	2.136700
25P	34015.480000
Mean	74455.814592
Median	70308.210000
50P	70308.210000
75P	107735.512500
Max	319210.270000
STD	52814.905897

```

In [23]: l=[]
for i in num:
    data=visa_df[i]
    count=len(data)
    Min=np.min(data)
    P_25=np.quantile(data,0.25) # 0 to 1
    Mean= np.mean(data)
    Median= np.median(data)
    P_50=np.quantile(data,0.5) # 0 to 1
    P_75=np.quantile(data,0.75) # 0 to 1
    Max=np.max(data)
    SD=np.std(data)
    l.append([count,Min,P_25,Mean,Median,P_50,P_75,Max,SD])

```

```

In [31]: idx=['Count','Min','25P','Mean','Median','50P','75P','Max','STD']
pd.DataFrame(l,
             columns=idx,
             index=num).T

```

```
Out[31]:
```

	no_of_employees	yr_of_estab	prevailing_wage
Count	25480.000000	25480.000000	25480.000000
Min	-26.000000	1800.000000	2.136700
25P	1022.000000	1976.000000	34015.480000
Mean	5667.043210	1979.409929	74455.814592
Median	2109.000000	1997.000000	70308.210000
50P	2109.000000	1997.000000	70308.210000
75P	3504.000000	2005.000000	107735.512500
Max	602069.000000	2016.000000	319210.270000
STD	22877.479904	42.366098	52814.905897

describe

```
In [33]: visa_df.describe()
```

```
Out[33]:
```

	no_of_employees	yr_of_estab	prevailing_wage
count	25480.000000	25480.000000	25480.000000
mean	5667.043210	1979.409929	74455.814592
std	22877.928848	42.366929	52815.942327
min	-26.000000	1800.000000	2.136700
25%	1022.000000	1976.000000	34015.480000
50%	2109.000000	1997.000000	70308.210000
75%	3504.000000	2005.000000	107735.512500
max	602069.000000	2016.000000	319210.270000

```
In [36]: # 2d Matric Transpose Columns<==>Rows
```

Percentiles

```
In [38]: P_25=np.percentile(wage_data,25)
P_25
```

```
Out[38]: 34015.479999999996
```

```
In [ ]: - 25percentage of total applicants having salary less than 34015
- 50percentage of total applicants having salary less than 70308.21
```

```
In [43]: wage_data=visa_df['prevailing_wage']
np.sum(wage_data<P_25)
```

```
Out[43]: 6370
```

```
In [ ]: 25percentage of total applicants having salary less than 34015
        6370             having salary less than 34015
```

```
In [45]: 25*25480/100
```

```
Out[45]: 6370.0
```

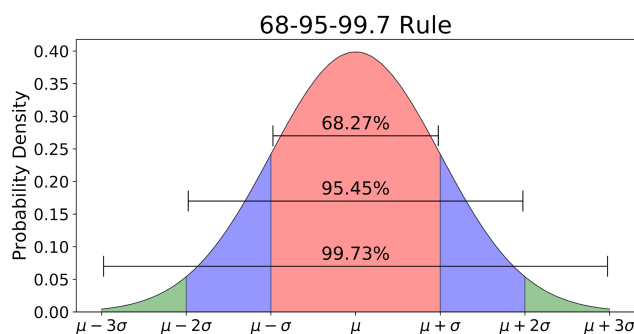
```
In [47]: wage_data=visa_df['prevailing_wage']
np.sum(wage_data<P_25)==25*25480/100
```

```
Out[47]: True
```

```
In [1]: # Import the packages
# read the data
# cat and num

import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns

visa_df=pd.read_csv(r'C:\Users\omkar\OneDrive\Documents\Gen_AI\Data_files\Visada
cat=visa_df.select_dtypes(include='object').columns
num=visa_df.select_dtypes(exclude='object').columns
```



Emperical Rule

- 68% of data coverage between **U-1Sigma to U+1Sigma**

```
In [7]: # step-1: Read the wage data
# step-2: calculate mean
# step-3: calculate std
# step-4: lb= s2-1*s3
# step-5: ub = s2+1*s3
# step-6: con1= wage_data>lb
# step-7: con2= wage_data<ub
# step-8: con=con1 & con2
# step-9: count=np ===== 100
# step-10: visa_df[con] data
# step-11: 68%(25480) = 100
```

```
In [15]: wage_data=visa_df['prevailing_wage']
wage_mean=np.mean(wage_data)
wage_std=np.std(wage_data)
```

```

lb=wage_mean-1*wage_std
ub=wage_mean+1*wage_std
con1= wage_data>lb
con2= wage_data<ub
con=con1 & con2
count=np.sum(con)
np.sum(con)==68*(25480)/100

```

Out[15]: False

```

In [25]: def emperical(num,alpha):
          wage_data=visa_df['prevailing_wage']
          wage_mean=np.mean(wage_data)
          wage_std=np.std(wage_data)
          lb=wage_mean-alpha*wage_std
          ub=wage_mean+alpha*wage_std
          con1= wage_data>lb
          con2= wage_data<ub
          con=con1 & con2
          count=np.sum(con)
          print('actual count:',count)
          print(f'{num} count:',num*(25480)/100)
          return(np.sum(con)==num*(25480)/100)
          emperical(68,1),emperical(95,2),emperical(99.7,3)

```

```

actual count: 17170
68 count: 17326.4
actual count: 24582
95 count: 24206.0
actual count: 25186
99.7 count: 25403.56

```

Out[25]: (False, False, False)

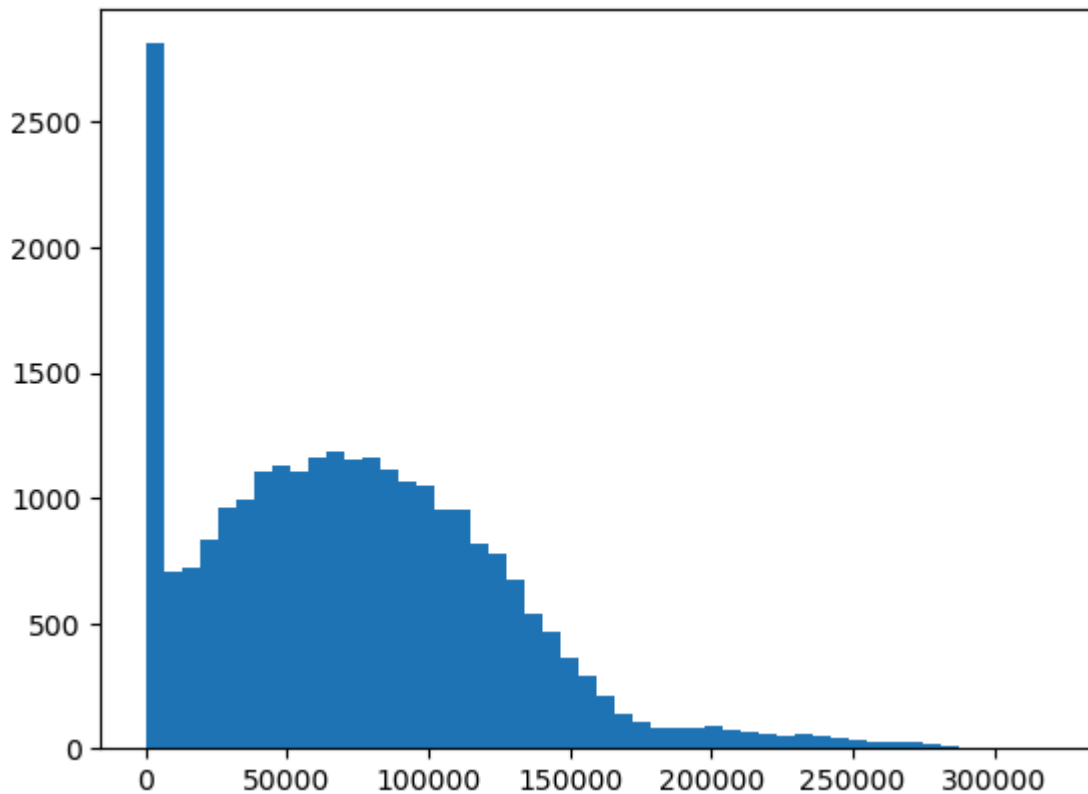
the above analysis conclude data does not follow normality

histogram

```

In [39]: count,intervals,n=plt.hist(wage_data,bins=50)

```



In [41]: count

```
Out[41]: array([2813., 708., 725., 832., 960., 993., 1109., 1132., 1108.,
1162., 1184., 1154., 1159., 1117., 1067., 1047., 953., 955.,
816., 780., 673., 539., 469., 364., 289., 211., 139.,
107., 81., 86., 79., 87., 78., 69., 60., 49.,
57., 52., 45., 37., 28., 27., 26., 20., 13.,
4., 5., 4., 4., 4.])
```

In [43]: intervals

```
Out[43]: array([2.13670000e+00, 6.38629937e+03, 1.27704620e+04, 1.91546247e+04,
2.55387874e+04, 3.19229500e+04, 3.83071127e+04, 4.46912754e+04,
5.10754380e+04, 5.74596007e+04, 6.38437634e+04, 7.02279260e+04,
7.66120887e+04, 8.29962514e+04, 8.93804140e+04, 9.57645767e+04,
1.02148739e+05, 1.08532902e+05, 1.14917065e+05, 1.21301227e+05,
1.27685390e+05, 1.34069553e+05, 1.40453715e+05, 1.46837878e+05,
1.53222041e+05, 1.59606203e+05, 1.65990366e+05, 1.72374529e+05,
1.78758691e+05, 1.85142854e+05, 1.91527017e+05, 1.97911179e+05,
2.04295342e+05, 2.10679505e+05, 2.17063667e+05, 2.23447830e+05,
2.29831993e+05, 2.36216155e+05, 2.42600318e+05, 2.48984481e+05,
2.55368643e+05, 2.61752806e+05, 2.68136969e+05, 2.74521131e+05,
2.80905294e+05, 2.87289457e+05, 2.93673619e+05, 3.00057782e+05,
3.06441945e+05, 3.12826107e+05, 3.19210270e+05])
```

In [45]: n

Out[45]: <BarContainer object of 50 artists>

interval	count
2.13670000e+00-6.38629937e+03	2813
6.38629937e+03-1.27704620e+04	708

```
In [49]: # wage = 2.13 to wage= 6386.2 there are 2813
lb=2.13670000e+00
ub=6.38629937e+03
con1=wage_data>=lb
con2=wage_data<=ub
con=con1 & con2
visa_df[con] # extraction
np.sum(con) # number
```

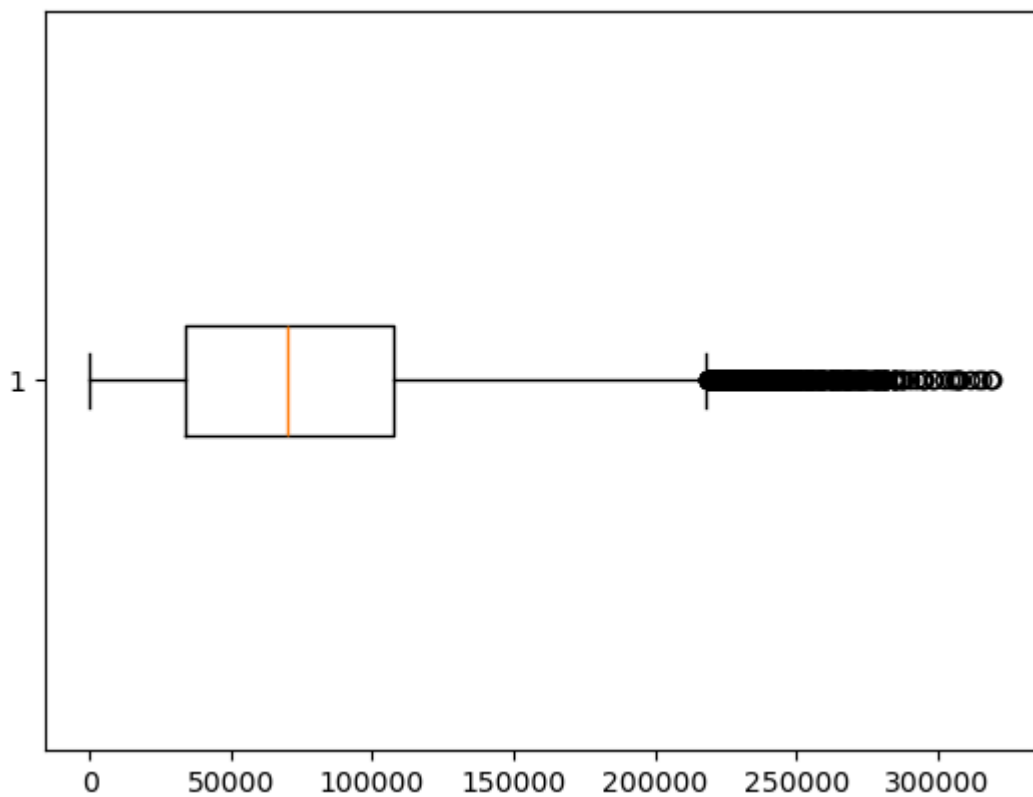
```
# con1= wage_data>lb
# con2= wage_data<ub
# con=con1 & con2
# count=np.sum(con)
```

Out[49]: 2813

outlier analysis

```
In [ ]:      Q1-1.5IQR  Q1  median  Q3  Q3+1.5IQR
            o      |-----|-----|-----|-----| o o
            flier  |-----|-----|-----|-----| fliers
                   <----->
                   IQR
```

```
In [58]: plt.boxplot(wage_data,vert=False)
plt.show()
```



```
In [1]: import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
```

```
visa_df=pd.read_csv(r'C:\Users\omkar\OneDrive\Documents\Gen_AI\Data_files\Visada
cat=visa_df.select_dtypes(include='object').columns
num=visa_df.select_dtypes(exclude='object').columns
```

```
In [12]: wage_data=visa_df['prevailing_wage']
q1=np.quantile(wage_data,0.25)
q2=np.quantile(wage_data,0.5)
q3=np.quantile(wage_data,0.75)
IQR=q3-q1
lb=q1-1.5*IQR
ub=q3+1.5*IQR
con1=wage_data<lb
con2=wage_data>ub
con=con1|con2
np.sum(con)
outliers_df=visa_df[con]
outliers_df
```

```
Out[12]:
```

	case_id	continent	education_of_employee	has_job_experience	requires_job_1
--	---------	-----------	-----------------------	--------------------	----------------

14	EZYV15	Asia	Master's	Y	
34	EZYV35	Asia	Master's	N	
130	EZYV131	South America	High School	N	
216	EZYV217	Asia	Master's	Y	
221	EZYV222	North America	Doctorate	Y	
...	...	...	...	...	
25191	EZYV25192	Asia	Master's	N	
25195	EZYV25196	North America	Master's	Y	
25468	EZYV25469	Asia	Bachelor's	N	
25469	EZYV25470	North America	Master's	Y	
25476	EZYV25477	Asia	High School	Y	

427 rows × 12 columns



```
In [14]: wage_data=visa_df['prevailing_wage']
q1=np.quantile(wage_data,0.25)
q2=np.quantile(wage_data,0.5)
q3=np.quantile(wage_data,0.75)
IQR=q3-q1
lb=q1-1.5*IQR
ub=q3+1.5*IQR
con1=wage_data>lb
con2=wage_data<ub
con=con1&con2
np.sum(con)
```

```
non_outliers_df=visa_df[con]
len(non_outliers_df)
```

Out[14]: 25053

```
In [9]: 25053+427
```

Out[9]: 25480

Imputaion of Outliers

```
In [ ]: visa_df # 25480
outliers_df # 427
non_outliers_df # 25053
```

```
In [18]: wage_data=visa_df['prevailing_wage'] # 25480
wage_data.values # this array has outliers

# we need to call each value
# bro are you available in outliers_df['prewage']
```

Out[18]: array([592.2029, 83425.65 , 122996.86 , ..., 146298.85 ,
86154.77 , 70876.91])

```
In [32]: len(outliers_df['prevailing_wage'])
```

Out[32]: 427

```
In [46]: outliers_df['prevailing_wage'].values
```

```
Out[46]: array([220081.73, 225569.73, 247393.01, 269321.68, 219529.62, 232227.33,
 238691.32, 220448.17, 230750.48, 235339.91, 232680.65, 256261.78,
 262189. , 250510.67, 218554.78, 256205.38, 221944.22, 229950.7 ,
 222628.84, 229819.69, 233641.72, 274019.43, 218982.83, 230984.28,
 223128.23, 246705. , 240266.34, 226090.72, 231949.27, 254604.08,
 288318.91, 244457.48, 242146.48, 247545.23, 234139.17, 231834.72,
 222730.57, 239773.63, 220553.95, 237539.32, 266440.49, 277281.01,
 275627.59, 249291.12, 292106.59, 234012.11, 247009.24, 267726.09,
 222098.09, 280661.13, 277984.52, 262852.71, 263877.21, 230916.64,
 264760.64, 267868.56, 222221.67, 242406.32, 276894.08, 281832.93,
 221033.57, 258292.84, 223492.85, 245418.67, 250289.08, 271344.69,
 252825.11, 230843.7 , 253668.86, 245819.17, 239465.3 , 236316.36,
 225268.13, 228941.33, 306021.96, 228074.25, 270799.93, 221982.69,
 256903.7 , 242813.07, 253111.05, 268440.33, 225681.78, 233150.41,
 307410.02, 227609.87, 248247.63, 244813.07, 233642.47, 246955.19,
 304543.67, 236918.29, 229494.55, 228805.91, 250899.07, 259739.35,
 240798.56, 238109.72, 281418.03, 239151.61, 280205.27, 272756.53,
 298520.02, 242018.37, 250436.94, 254275.73, 254230.21, 237715.74,
 235224.73, 232265.54, 259277.18, 271409.44, 232792.93, 259097.55,
 249339.52, 238893.91, 234406.36, 253915.96, 238078.87, 231936.29,
 222457.51, 255626.72, 247602.36, 227360.23, 284328.96, 219351.89,
 248137.05, 273774.05, 246806.65, 225816.56, 277357.78, 238353.7 ,
 254858.86, 234856.36, 226588.62, 247935.5 , 226927.95, 225181.88,
 240846.68, 240467.47, 248608.01, 261729.62, 227035.49, 278629.2 ,
 236523.12, 222416.49, 240193.11, 243086.94, 294258.8 , 224366.03,
 254361.39, 247003.95, 244758.73, 284585.2 , 260321.86, 239400.63,
 229535.14, 246525.47, 269645.3 , 228937.44, 228767.83, 245999.3 ,
 233767.78, 244938.45, 267448.28, 245110.1 , 264564.71, 219128.94,
 221860.43, 262556.25, 252003.71, 262314.42, 251730.64, 247876.09,
 223185.07, 258044.72, 278708.4 , 264362.95, 258477.93, 272311.21,
 289878.68, 227441.23, 245197.29, 248076.5 , 266110.44, 247525.17,
 219090.04, 222793.7 , 230919.67, 228715.49, 250163.1 , 268713.5 ,
 268737.76, 278409.92, 223098.15, 261496.17, 239894.57, 306206.67,
 222276.55, 273009.56, 238662.95, 253698.68, 263466.82, 218318.76,
 280175.95, 248668.8 , 234902.2 , 249626.02, 231138.15, 264798.08,
 234755.9 , 230566.11, 235429.79, 277104.8 , 240410.79, 315497.6 ,
 238065.59, 229565.95, 233013.3 , 229794.44, 220817.57, 228734.21,
 256618.11, 226892.03, 243111.61, 251014.23, 230922.38, 248492.44,
 232673.86, 257966.97, 233282.32, 225877.49, 259461.84, 255674.94,
 259083.16, 250841.05, 261556.56, 219465.03, 273951.68, 294807.32,
 220736.86, 265168.9 , 263908.89, 278554.98, 239880.69, 254861.76,
 281582.84, 224049.27, 238039.66, 238165.9 , 268110.52, 228645.15,
 257781.92, 219284.2 , 318446.05, 307716.1 , 273426.75, 249850.51,
 286336.94, 248533.33, 259835.69, 261554.61, 242597.19, 226669.55,
 306982.28, 269990.25, 244202.66, 233790.85, 281590.41, 224874.12,
 230017.17, 221735.9 , 271877.51, 257916.69, 314156.06, 229874.06,
 235694.88, 280155.01, 272286.3 , 266071.13, 237874.32, 251654.81,
 280521.57, 263400. , 224151.28, 275926.27, 273025.14, 227562.89,
 229208.81, 284220.42, 246099.93, 219321.41, 221669.11, 230395.29,
 221900.52, 245785.16, 294945.51, 236692.29, 236184.68, 249610.99,
 257775.44, 239224.53, 221524.92, 254015.02, 281344.22, 297589.88,
 221595.07, 226169.32, 287433.9 , 220280.21, 252808.74, 262777.69,
 268204.62, 257994.73, 302513.77, 222511.73, 256502.28, 245599.1 ,
 224523.8 , 239441.23, 259556.53, 245890.1 , 233990.98, 240153.33,
 221588.52, 223357.39, 266821.59, 247248.66, 230431.69, 240099.97,
 255771.26, 247560.19, 267189.12, 234330.77, 287191.2 , 219056.42,
 232389.85, 221498.71, 266813.88, 232473.44, 222868.81, 241680.07,
 284205.22, 241811.15, 240301.92, 254522.41, 221354.25, 247084.9 ,
 238778.59, 223837.82, 219968.65, 232462.72, 311734.49, 319210.27,
 225324.61, 242673.44, 242507.42, 244049.04, 274436.62, 240609.17,
```

```

231149.93, 218826. , 255033.75, 228520.51, 285088.16, 221058.13,
235454.33, 285256.54, 222177.4 , 232933.22, 239617.43, 233585.11,
249452.91, 227133.68, 239600.76, 242466.81, 225057.61, 278250.32,
241928.36, 250319.73, 225036.07, 249907.87, 252692.71, 264947.49,
223062.79, 229456.89, 249394.83, 238788.29, 225440.67, 221757.19,
218823.72, 240922.83, 232029.62, 254157.54, 270184.42, 238178.7 ,
227683.38, 270599.15, 232111.27, 264303.76, 243190.47, 226982.69,
230428.5 , 275211.23, 232068.71, 231426.9 , 223673.34, 252661.63,
242247.08, 235875.97, 229568.14, 247355.35, 265969.8 , 231545.19,
232467.06, 247626.74, 242347.14, 220664.53, 248375.16, 218488.72,
236625.64, 268376.5 , 280482.51, 234308.77, 272715.74, 273772.47,
279174.79])

```

```

In [36]: wage_median=visa_df['prevailing_wage'].median()
wage_median

```

Out[36]: 70308.20999999999

```

In [38]: wage_median=visa_df['prevailing_wage'].median()
l=[]
for i in visa_df['prevailing_wage'].values:
    if i in outliers_df['prevailing_wage'].values:
        l.append(wage_median)
    else:
        l.append(i)
max(l)

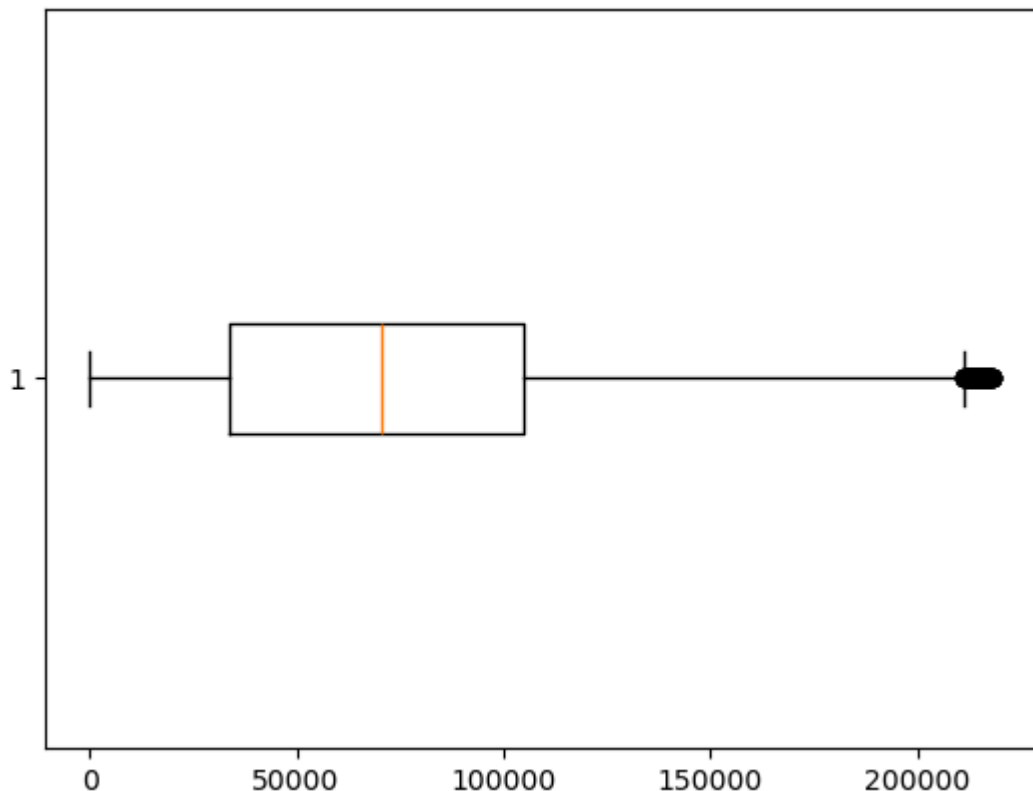
```

Out[38]: 218094.61

```

In [42]: plt.boxplot(l,vert=False)
plt.show()

```



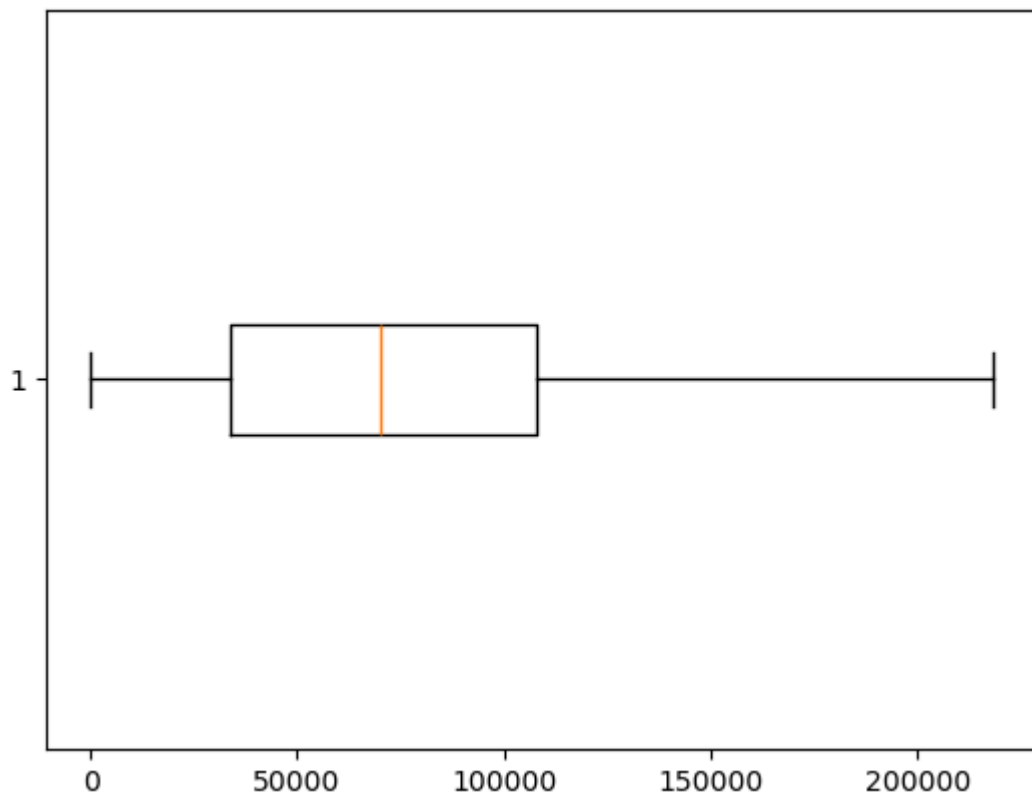
clip method or winsorization

- we will fill less than lb values with lb
- we will fill greater than ub values with ub

```
In [52]: print('lb:',lb)
print('ub:',ub)
new_wage_data=wage_data.clip(lb,ub)
# wage_data = 25480
#           25053+427
```

```
lb: -76564.56875000002
ub: 218315.56125000003
```

```
In [54]: plt.boxplot(new_wage_data,vert=False)
plt.show()
```

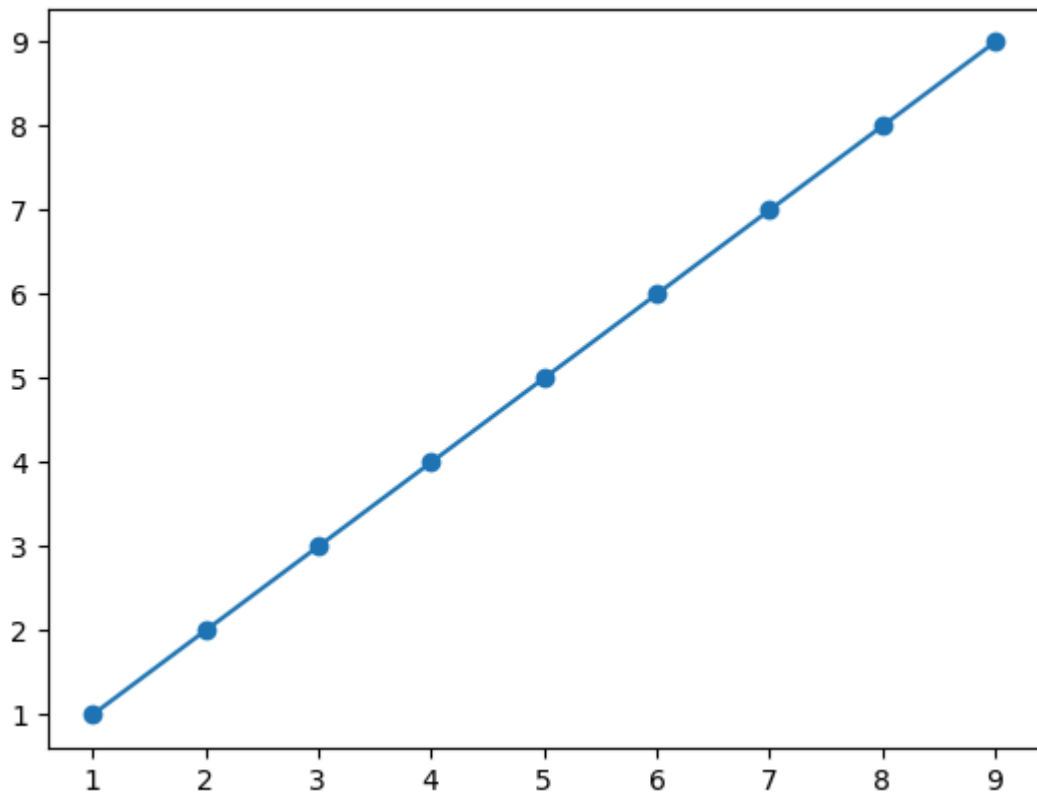


Scatter plots

- A plot between **numerical vs numerical**

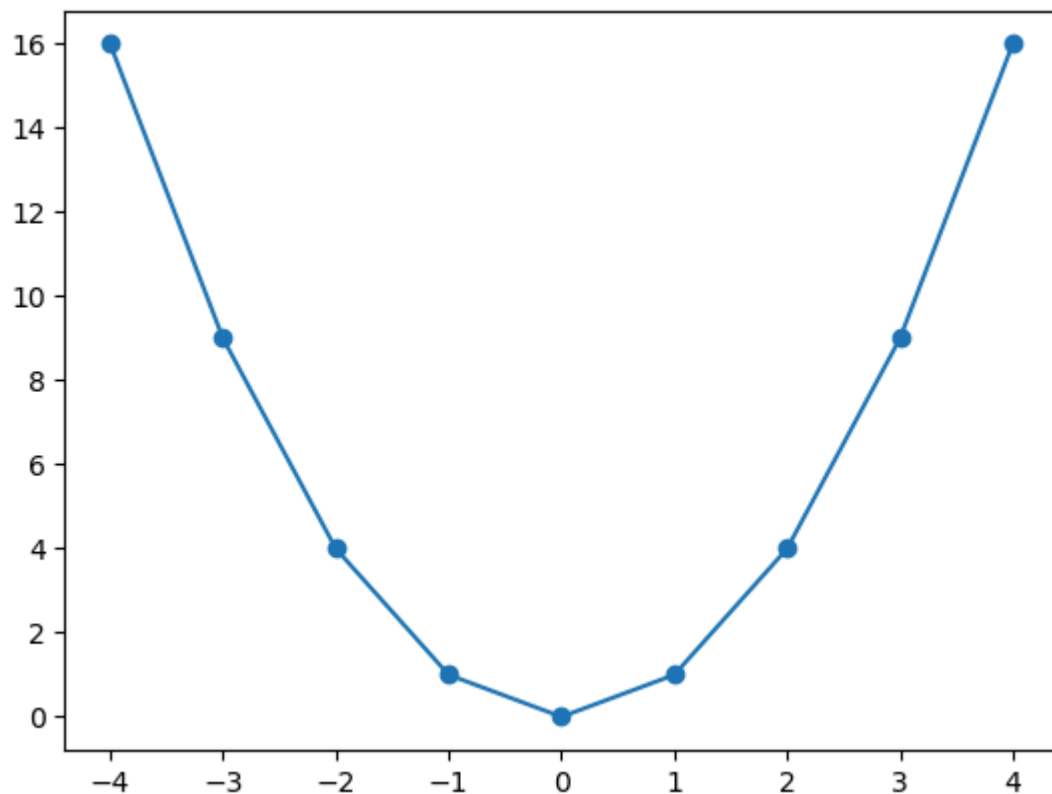
```
In [60]: l1=range(1,10)
l2=range(1,10)
plt.scatter(l1,l2)
plt.plot(l1,l2)
```

```
Out[60]: [<matplotlib.lines.Line2D at 0x2000733d550>]
```



```
In [64]: # draw the parabola
l1=[-4,-3,-2,-1,0,1,2,3,4]
l2=[16,9,4,1,0,1,4,9,16]
plt.scatter(l1,l2)
plt.plot(l1,l2)
```

```
Out[64]: [<matplotlib.lines.Line2D at 0x20003bd78c0>]
```



```
In [ ]: # Import the packages
# read the data
```

```
# Convert cat and numerical variables
```

```
In [4]: import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns

visa_df=pd.read_csv(r'C:\Users\omkar\OneDrive\Documents\Gen_AI\Data_files\Visada
cat=visa_df.select_dtypes(include='object').columns
num=visa_df.select_dtypes(exclude='object').columns
```

task

```
In [6]: cat
```

```
Out[6]: Index(['case_id', 'continent', 'education_of_employee', 'has_job_experience',
              'requires_job_training', 'region_of_employment', 'unit_of_wage',
              'full_time_position', 'case_status'],
              dtype='object')
```

```
In [10]: # 'continent' : "there are 6 continents are there"
# 'case_status' : 'Certified and Denied'
```

```
In [20]: visa_df['case_status'].value_counts()
```

```
Out[20]: case_status
Certified    17018
Denied       8462
Name: count, dtype: int64
```

```
In [22]: # 17018 = 6 continents
# 8462 = 6 continents
```

```
In [24]: visa_df['continent'].value_counts()
```

```
Out[24]: continent
Asia          16861
Europe         3732
North America  3292
South America   852
Africa          551
Oceania         192
Name: count, dtype: int64
```

```
In [26]: # 16861 ppl are from asia applied visa= (Cer+De)
```

```
In [ ]: # How many ppl are certified from Asia
# How many ppl are debied from Asia
```

```
In [34]: con1=visa_df['continent']=='Asia'    # T
con2=visa_df['case_status']=='Certified' # T
con=con1&con2
np.sum(con)
visa_df[con]
```

```
Out[34]: 11012
```

```
In [ ]: 11012 applicants from asia are certified
```

```
In [40]: con1=visa_df['continent']=='Asia' # T
con2=visa_df['case_status']=='Denied' # T
con=con1&con2
np.sum(con)
visa_df[con]
```

```
Out[40]:
```

	case_id	continent	education_of_employee	has_job_experience	requires_job_1
0	EZYV01	Asia	High School	N	
2	EZYV03	Asia	Bachelor's	N	
3	EZYV04	Asia	Bachelor's	N	
11	EZYV12	Asia	High School	Y	
15	EZYV16	Asia	High School	Y	
...	...	...	...	...	...
25450	EZYV25451	Asia	Bachelor's	N	
25452	EZYV25453	Asia	High School	N	
25454	EZYV25455	Asia	Bachelor's	N	
25459	EZYV25460	Asia	High School	Y	
25461	EZYV25462	Asia	Master's	Y	

5849 rows × 12 columns

```
In [52]: for i in cat[1:]:
print(visa_df[i].unique())
```

```
['Asia' 'Africa' 'North America' 'Europe' 'South America' 'Oceania']
['High School' 'Master's' 'Bachelor's' 'Doctorate']
['N' 'Y']
['N' 'Y']
['West' 'Northeast' 'South' 'Midwest' 'Island']
['Hour' 'Year' 'Week' 'Month']
['Y' 'N']
['Denied' 'Certified']
```

```
In [58]: Certi_count,Deni_count=[],[]
Continent_labels=visa_df['continent'].unique()
for i in Continent_labels:
    con1=visa_df['continent']==i # T
    con2=visa_df['case_status']=='Certified' # T
    con3=visa_df['case_status']=='Denied'
    certi_con=con1&con2
    deni_con=con1&con3
    Certi_count.append(np.sum(certi_con))
    Deni_count.append(np.sum(deni_con))
```

```
In [112... df=pd.DataFrame()
df['Continents']=Continent_labels
```

```
df['Certified']=Certi_count
df['Denied']=Deni_count
df
```

Out[112...]

	Continents	Certified	Denied
0	Asia	11012	5849
1	Africa	397	154
2	North America	2037	1255
3	Europe	2957	775
4	South America	493	359
5	Oceania	122	70

In [77]: visa_df.groupby('continent')

Out[77]: <pandas.core.groupby.generic.DataFrameGroupBy object at 0x000002014EC7A1B0>

In [81]: visa_df.groupby('continent').count()

Out[81]:

	case_id	education_of_employee	has_job_experience	requires_job_training	nc
--	---------	-----------------------	--------------------	-----------------------	----

continent					
Africa	551	551	551	551	
Asia	16861	16861	16861	16861	
Europe	3732	3732	3732	3732	
North America	3292	3292	3292	3292	
Oceania	192	192	192	192	
South America	852	852	852	852	



In [89]: visa_df
 visa_df.groupby('continent')
 list(visa_df.groupby('continent'))
 list(visa_df.groupby('continent'))[0]
 list(visa_df.groupby('continent'))[0][1]

Out[89]:

	case_id	continent	education_of_employee	has_job_experience	requires_job_1
4	EZYV05	Africa	Master's	Y	
18	EZYV19	Africa	Master's	Y	
74	EZYV75	Africa	Master's	Y	
194	EZYV195	Africa	Master's	Y	
242	EZYV243	Africa	Bachelor's	N	
...	...	...	...	...	...
25385	EZYV25386	Africa	Doctorate	Y	
25408	EZYV25409	Africa	Master's	Y	
25443	EZYV25444	Africa	Bachelor's	N	
25446	EZYV25447	Africa	Master's	N	
25474	EZYV25475	Africa	Doctorate	N	

551 rows × 12 columns



In [93]: `list(visa_df.groupby('continent'))`
`visa_df.groupby('continent')['prevailing_wage'].mean()`

Out[93]:

continent	
Africa	77425.923450
Asia	79543.021780
Europe	59842.925899
North America	68066.319257
Oceania	75994.276719
South America	60209.575314

Name: prevailing_wage, dtype: float64

In [101... `visa_df[visa_df['continent']=='Africa']['prevailing_wage'].mean()`
`visa_df[visa_df['continent']=='Asia']['prevailing_wage'].mean()`

Out[101... 79543.02178043414

In [109... `# con1=visa_df['case_status']=='Certified'`
`# visa_df.groupby('continent')[].count()`

pd.crosstab

In [115... `# pd.crosstab(index, columns)`
`idx=visa_df['continent']`
`cols=visa_df['case_status']`
`pd.crosstab(idx,cols)`

Out[115...

	case_status	Certified	Denied
continent			
Africa		397	154
Asia		11012	5849
Europe		2957	775
North America		2037	1255
Oceania		122	70
South America		493	359

In [117...

```
idx=visa_df['continent']
col1=visa_df['case_status']
col2=visa_df['education_of_employee']
cols=[col1,col2]
pd.crosstab(idx,cols)
```

Out[117...

case_status				Certified			
education_of_employee	Bachelor's	Doctorate	High School	Master's	Bachelor's	Doctorate	
continent							
Africa	81	43	23	250	62	11	
Asia	4407	780	676	5149	2761	143	
Europe	1040	788	162	967	259	58	
North America	641	207	210	979	584	51	
Oceania	38	19	19	46	28	3	
South America	160	75	74	184	173	14	



numerical vs numerical

In [5]: num

Out[5]: Index(['no_of_employees', 'yr_of_estab', 'prevailing_wage'], dtype='object')

In [7]: visa_df.corr(numeric_only=True)

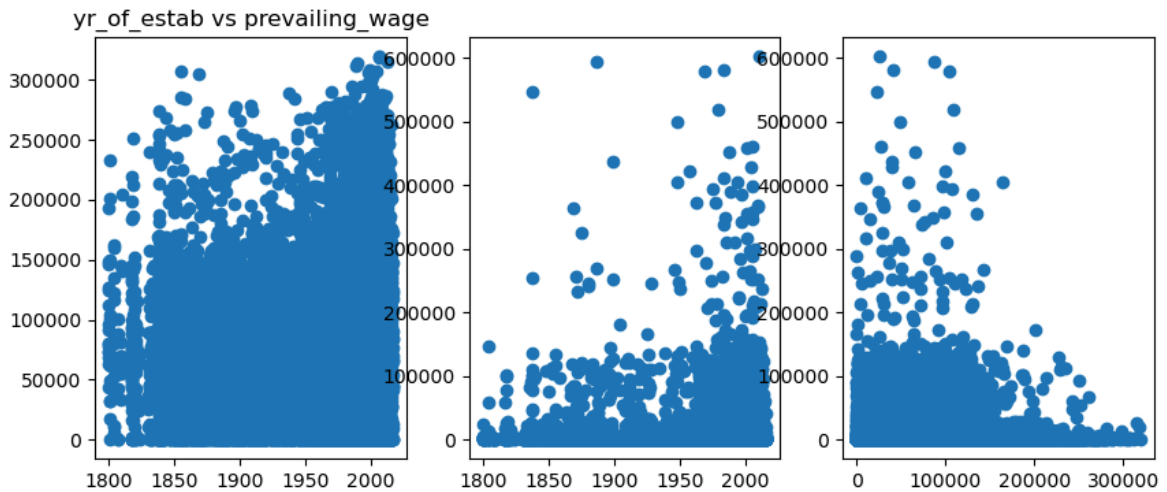
```
Out[7]:
```

	no_of_employees	yr_of_estab	prevailing_wage
no_of_employees	1.000000	-0.017770	-0.009523
yr_of_estab	-0.017770	1.000000	0.012342
prevailing_wage	-0.009523	0.012342	1.000000

```
In [9]: col1=visa_df['yr_of_estab']
col2=visa_df['prevailing_wage']
```

```
col3=visa_df['no_of_employees']
plt.figure(figsize=(10,4))
plt.subplot(1,3,1).scatter(col1,col2)
plt.title('yr_of_estab vs prevailing_wage')
plt.subplot(1,3,2).scatter(col1,col3)
plt.subplot(1,3,3).scatter(col2,col3)
```

Out[9]: <matplotlib.collections.PathCollection at 0x19d6ef58410>



```
In [ ]: # Google class room
#       Data Files
#       winequality data
# read the data
# its has all numerical columns
# correlation matrix
# 3 cases
#   highly postive case  c1 vs c2 scatter
#   highly negative case c1 vs c2 scatter
#   no relation         c1 vs c2 scatter
```

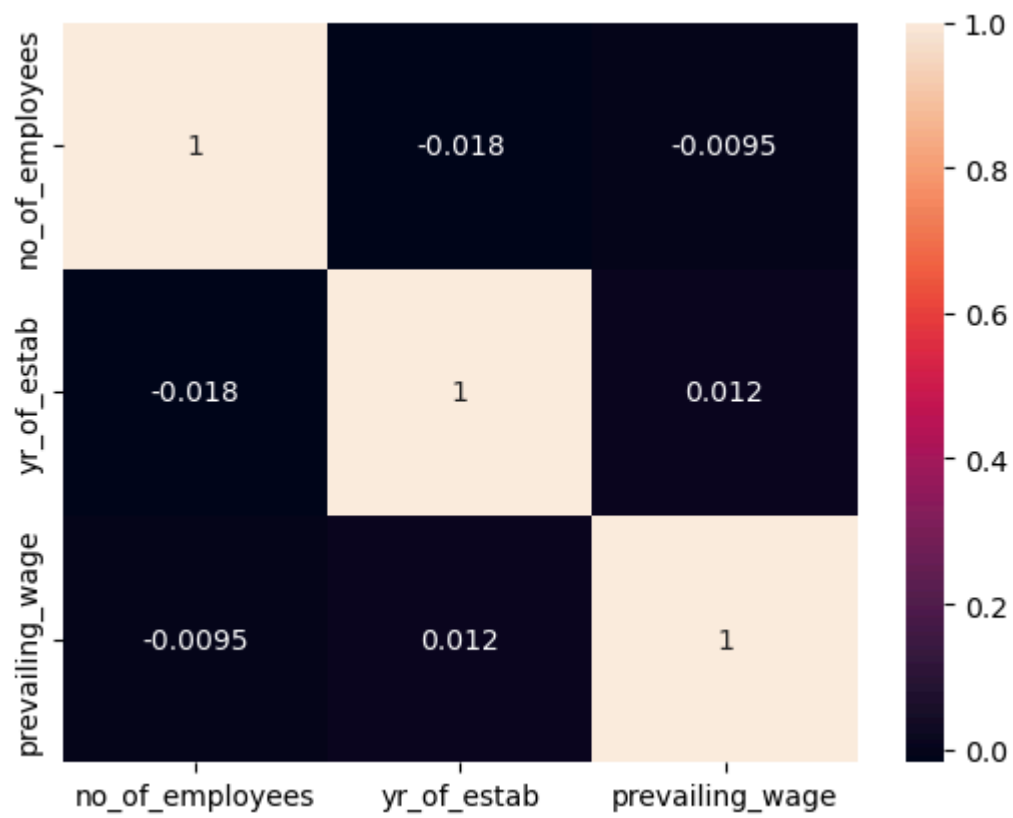
```
In [ ]: # when columns are huge
# so many values
# human eye can not able to figureit out very fast
# heatmap
```

heatmap

```
In [12]: corr=visa_df.corr(numeric_only=True)
```

```
In [16]: plt.figure()
sns.heatmap(corr,annot=True)
```

Out[16]: <Axes: >



```
In [ ]: bar
        pie
        histogram
        boxplot
        scatterplot
```

```
In [ ]:
```